

William Comess

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EXAMS

- Society of Actuaries Exam P – Passed July 2025
- Casualty Actuarial Society MAS-I – Expected November 2026

EDUCATION

University of Chicago

Chicago, IL

Bachelor of Arts (B.A.) in Mathematics and Statistics

Expected May 2027

- Cumulative GPA:** ???/4.00 | 2x AIME qualifier (2022 – 2023)
- Relevant Courses:** Machine Learning, Adv Numerical Analysis, Bayesian Statistics, and Linear Algebra

RESEARCH EXPERIENCE

Grossman Center for Quantitative Biology, UChicago

Chicago, IL

Student Researcher – Jaramillo Lab (Dept. of Neurobiology)

June 2026 – Present

- Developed optimal transport and Gaussian kernel smoothing interpolation algorithms to predict prefrontal cortex injection sites from thalamic trace data in R using the T4transport, transport, and sf packages
- Evaluated prediction accuracy via Leave-One-Out Cross-Validation and conducted goodness-of-fit tests showing residual error followed non-normal Gamma distributions
- Implemented computational neuroscience network models in Python by applying Euler's method to solve differential equations, while parallelizing R pipelines to optimize runtime across 3D point cloud simulations

Department of Mathematics, University of Notre Dame

Chicago, IL

Directed Study Mentee (Prof. Sam Evens)

June – August 2025

- Completed graduate-level readings and problem sets in group theory and ring theory, emphasizing homomorphisms, exact sequences, and structure theorems
- Studied modules over commutative rings, including submodules, quotient modules, and module homomorphisms

UChicago Department of Mathematics

Chicago, IL

REU Apprentice Program Participant

June – July 2024

- Wrote a paper on re-deriving the Central Limit Theorem (CLT) using both elementary statistics and measure theoretic probability with PhD student Marko Medvedev
- First Proved the CLT using moment-generating functions and elementary statistics, then re-proved using measure theory, Schwartz functions, and Fourier transforms
- Participated in weekly seminars on abstract algebra, random walks, the heat equation, and knot theory

Northwestern CIERA REACH

Evanston, IL

Student Researcher

June – July 2022

- Developed a model using NumPy, Pandas, and SciPy that estimates X-ray luminosity of accreting black holes in binary systems based on orbital period, mass transfer rate, and distance with astrophysicists
- Model showed a 15% improvement in accuracy over baseline predictions from existing literature

PROFESSIONAL EXPERIENCE

The University of Chicago, Computer Science Department

Chicago, IL

Tech Staff Student Worker

October 2023–July 2024

- Assisted with department-wide data migration of administrative data by updating internal documentation and standardizing metadata with hyperlinks through internal Confluence documentation, improving data integrity and accessibility with IT staff
- Provided on-demand technical and general support to an average of 10 users per week, resolving software and hardware issues efficiently and professionally

EXTRACURRICULAR INVOLVEMENTS

University of Chicago Data Science Society

Chicago, IL

Member and Project Participant

September 2025 – Present

- Cleaned and prepared dataset from the Card & Krueger minimum wage study
- Created Python pyplot visualizations replicating key findings on higher minimum wages on employment
- Built predictive models for FIFA World Cup outcomes using LASSO and Poisson regression

PROJECTS

Bayesian Econometric Housing Price Analysis (STAT 27410)

Chicago, IL

SKILLS

Technical Skills: Proficient in Microsoft Word, Excel, Overleaf, R, Python, Pandas, Pyplot, and Numpy

